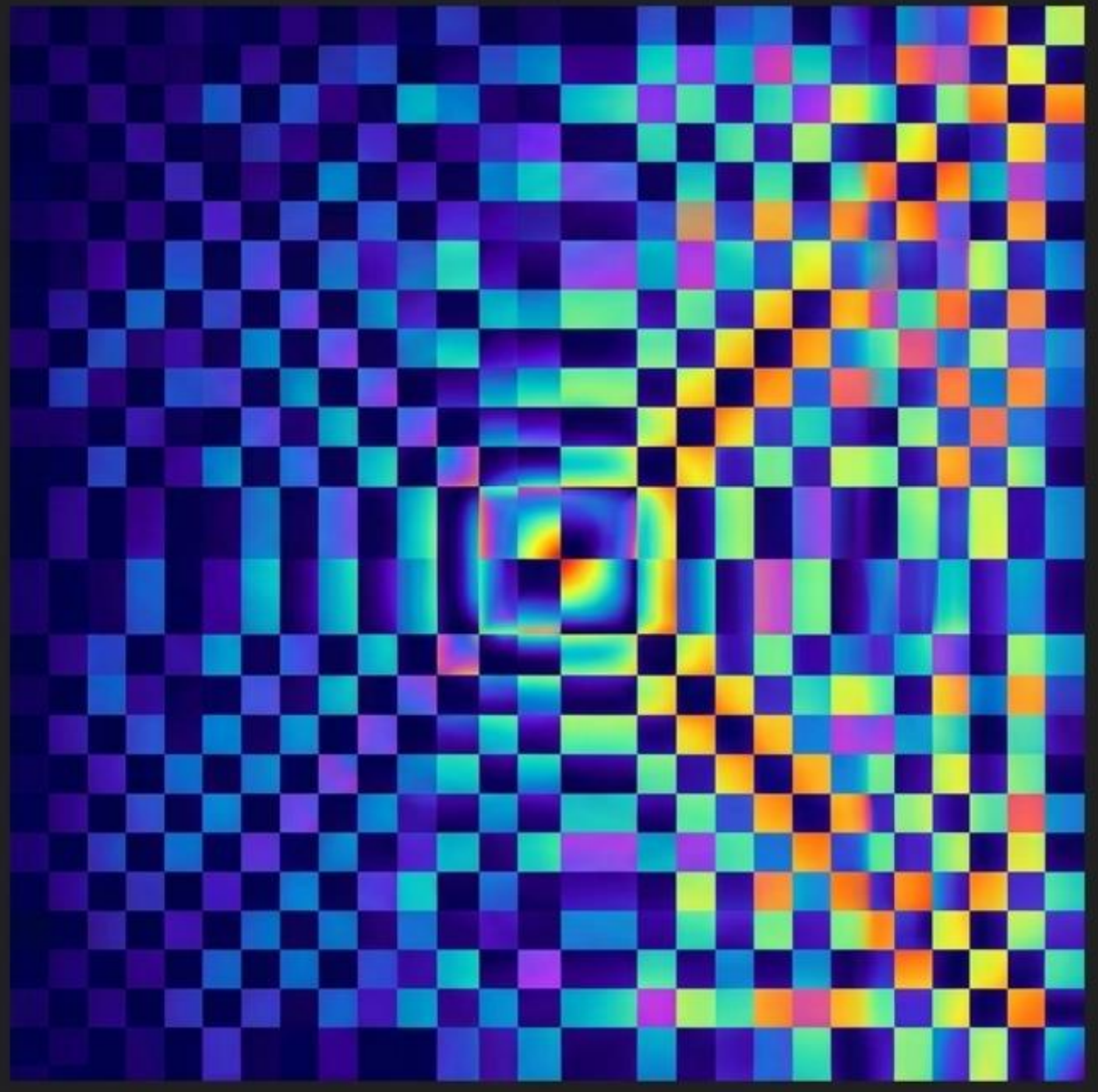
Seeing the Quantum Matrix

Visualizing QFT-NOT Compositions and X-Gate Mechanics with QAL



A Visual Proof Workbook | Experimental Mathematics in Action

Gemini Notebook

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & c_4 \\ a_{11} & a_{12} & a_{13} & c_4 \\ \vdots & \vdots & \vdots & \vdots \\ a_{21} & a_{22} & a_{23} & c_n \end{bmatrix}, \det(H - E^* I) = 0$$

$$|\psi\rangle = \sum_i c_i |\varphi_i\rangle$$

$$\psi = \begin{bmatrix} a_{ij1} + b_{21} \\ \det(H - E^* I) \end{bmatrix}$$

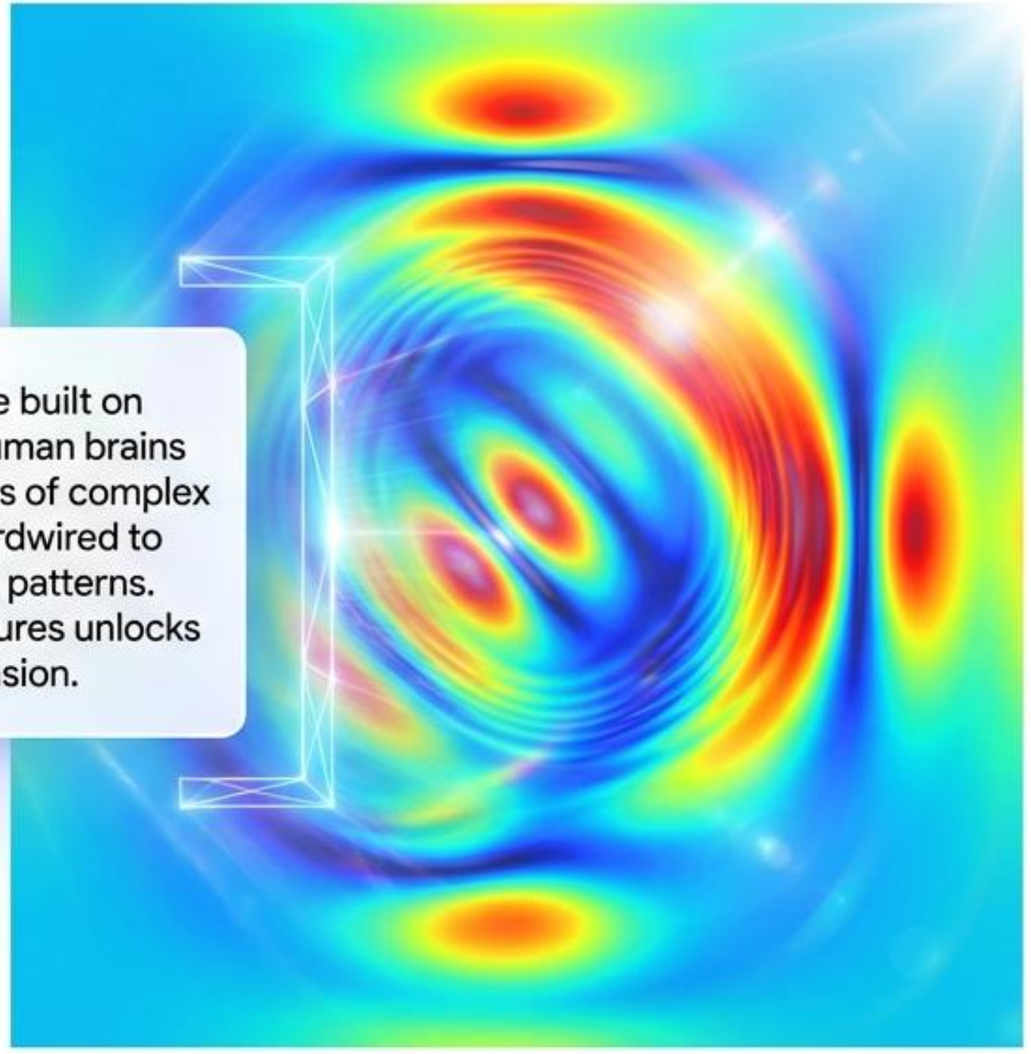
$$\langle \phi_j | A | \phi_i \rangle \quad k = \frac{}{k}$$

$$A = [a_{ij}], \quad \det(H - E^* I) = 0$$

$$\det(H - E^* I) = 0 \quad |\psi\rangle = \sum c_i |\phi_i\rangle$$

$$\langle \phi_j | A | \phi_i \rangle$$

Quantum algorithms are built on complex linear algebra. Human brains aren't wired to read matrices of complex numbers—but we are hardwired to understand pictures and patterns. Harnessing interactive pictures unlocks intuitive comprehension.

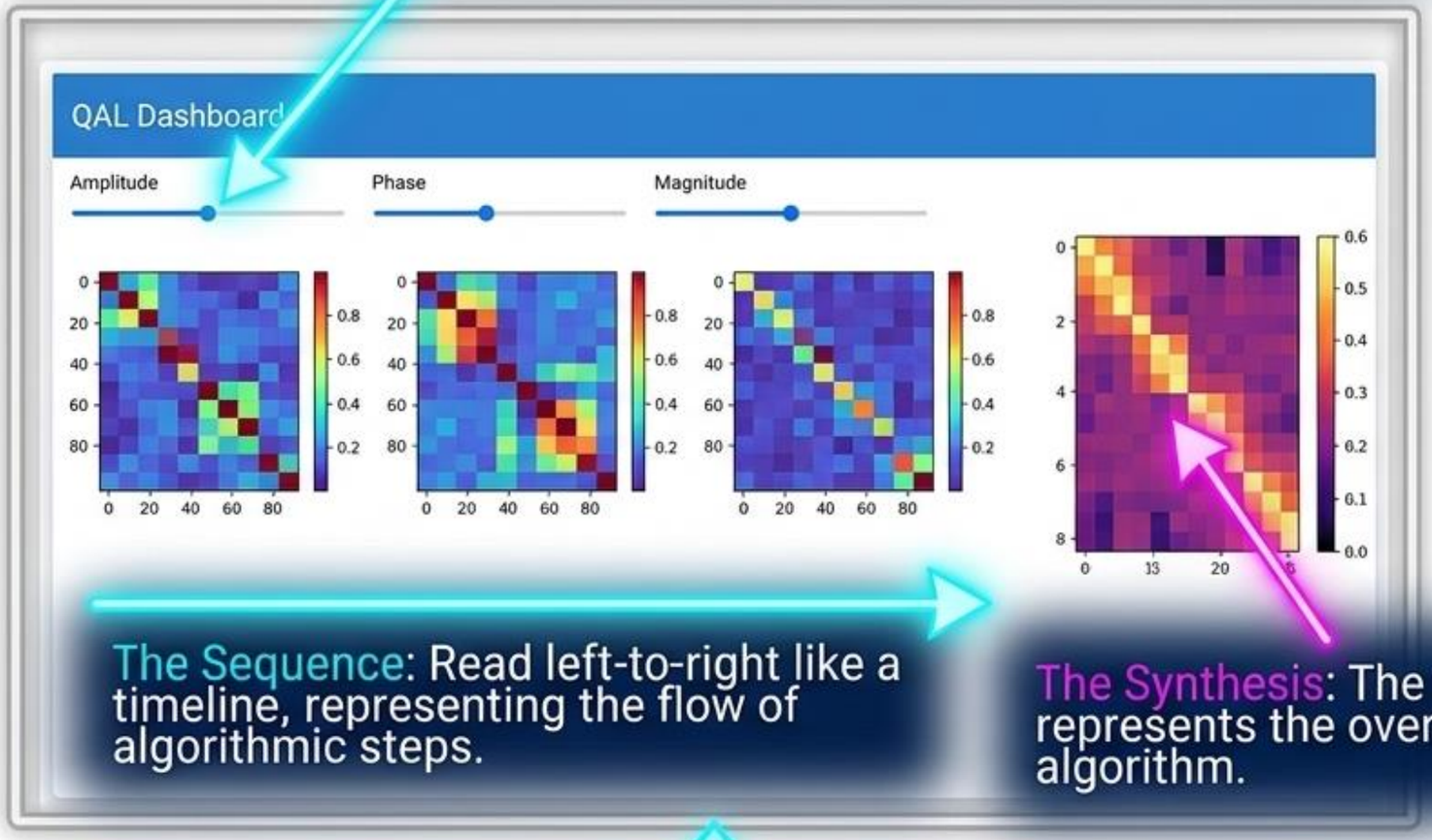


Traditional Mathematical Modeling

Visual Interactive Mathematics

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The Dashboard: Direct interaction without typing; sliders modify complex amplitudes in real-time.



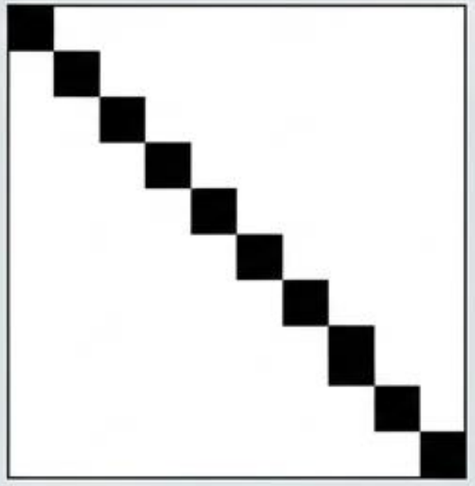
The Sequence: Read left-to-right like a timeline, representing the flow of algorithmic steps.

The Synthesis: The final heatmap represents the overall operation of the algorithm.

QAL translates opaque algorithmic steps into a sequence of color-coded matrices. Hue represents phase; saturation represents magnitude.

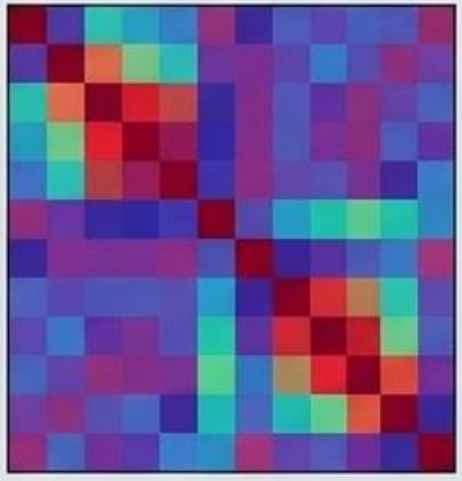
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Specimen A: The X Gate (Logical NOT)



A simple permutation matrix. An anti-diagonal line of black squares representing a binary flip.

Specimen B: Quantum Fourier Transform

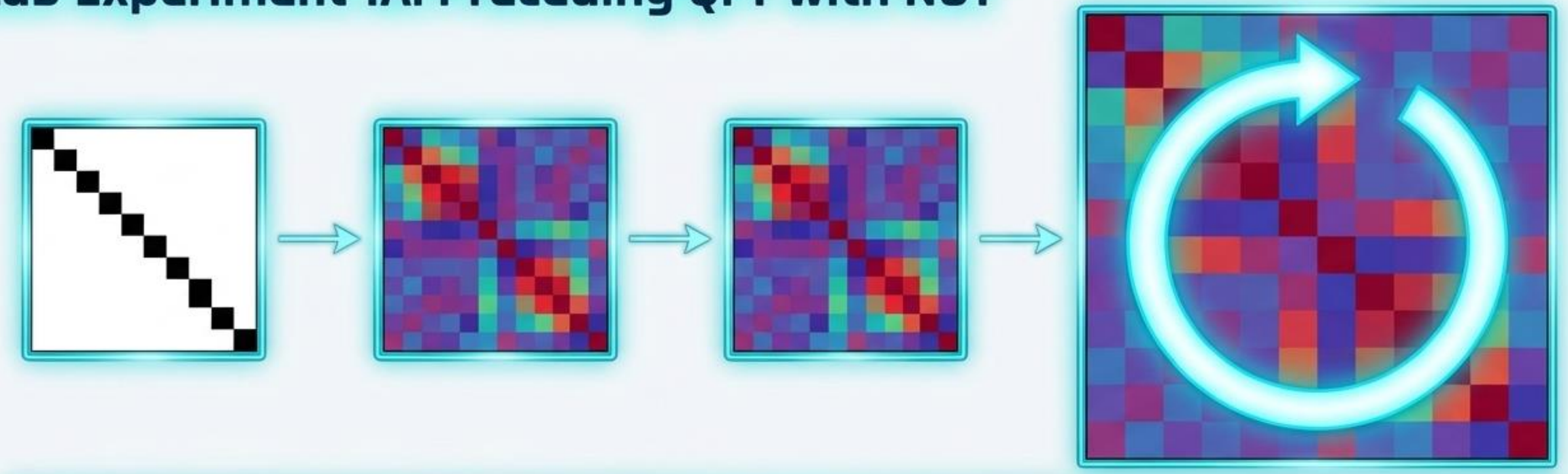


A complex operation that alters quantum phases, represented by a multidirectional continuum of colors.

The Hypothesis: What happens to the quantum state when we compose a simple permutation with a complex phase transformation?

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Lab Experiment 1A: Preceding QFT with NOT



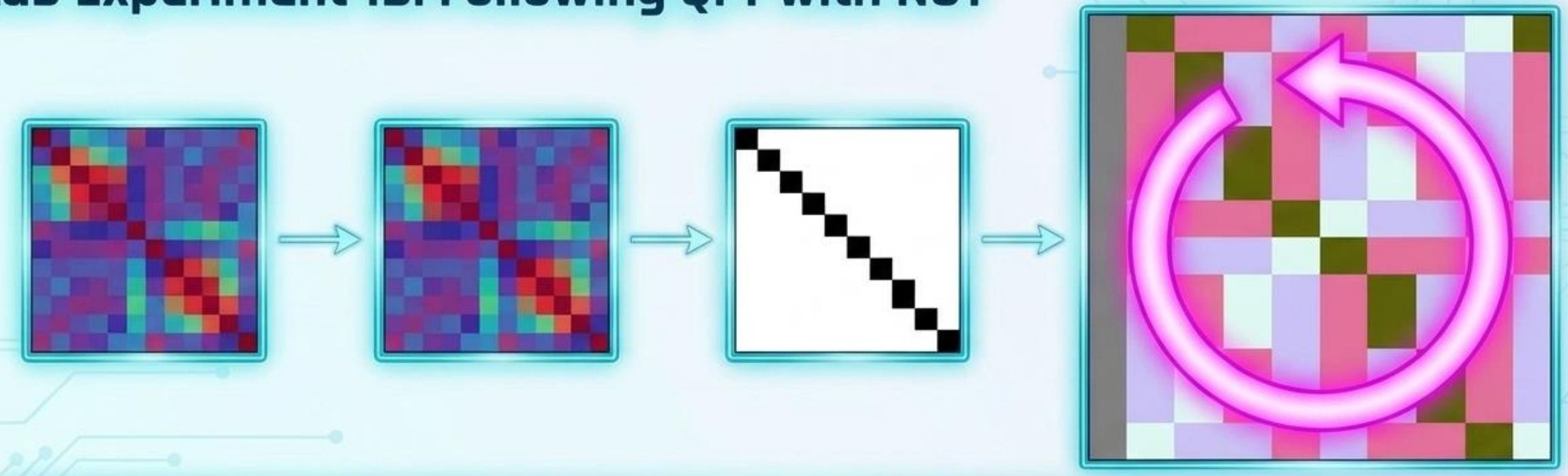
The Equation: $QFT \circ QFT \circ X$

The Observation: When the X gate (NOT) is applied before two applications of QFT, the resulting overall operation represents 'Add 1' with wraparound.

The Visual Truth: This mathematically translates to a Successor operation, visually manifested as a distinct Clockwise Rotation in the matrix space.

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Lab Experiment 1B: Following QFT with NOT

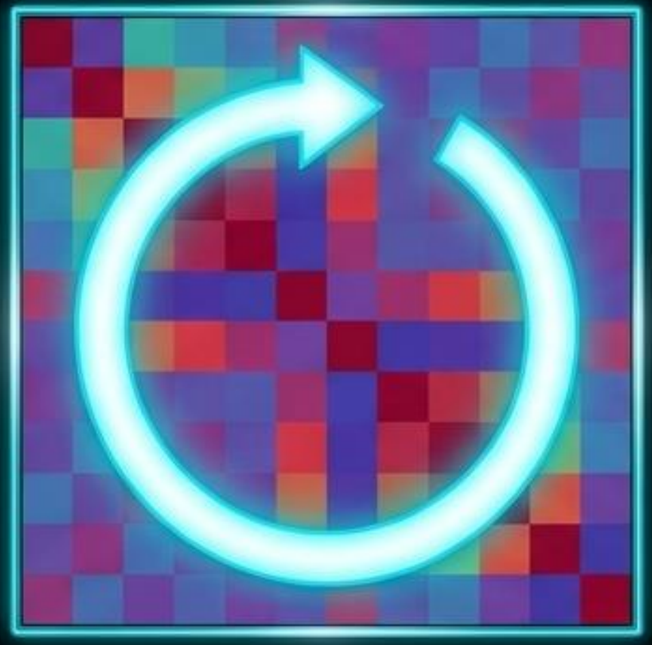
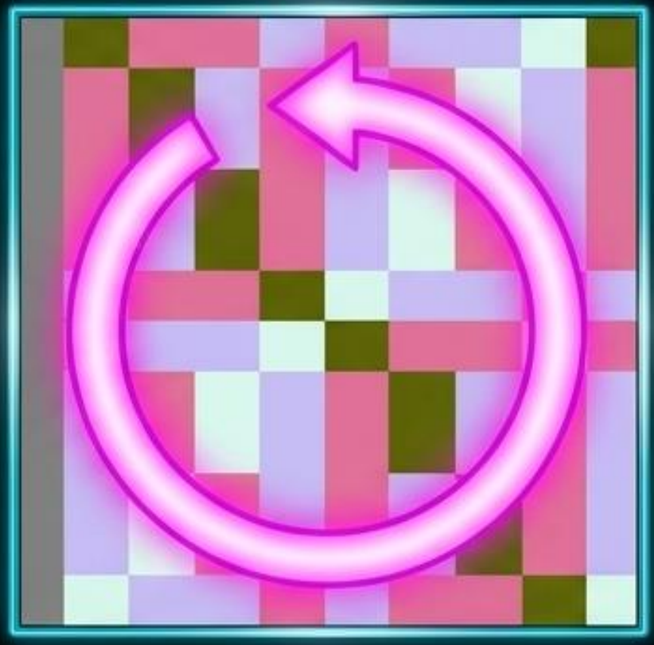


The Equation: $X \circ QFT \circ QFT$

The Observation: When the X gate is applied after two applications of QFT, the resulting overall operation becomes a 'Subtract 1'.

The Visual Truth: This mathematically translates to a Predecessor operation, visually manifested as a Counter-Clockwise Rotation.

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$QFT^2 \circ X$	$X \circ QFT^2$
	
Add 1 / Clockwise	Subtract 1 / Counter-Clockwise

The Power of Sequence. Order matters. Merely moving the X gate from the beginning of the algorithm to the end entirely reverses the rotational direction of the resulting quantum state. But why?

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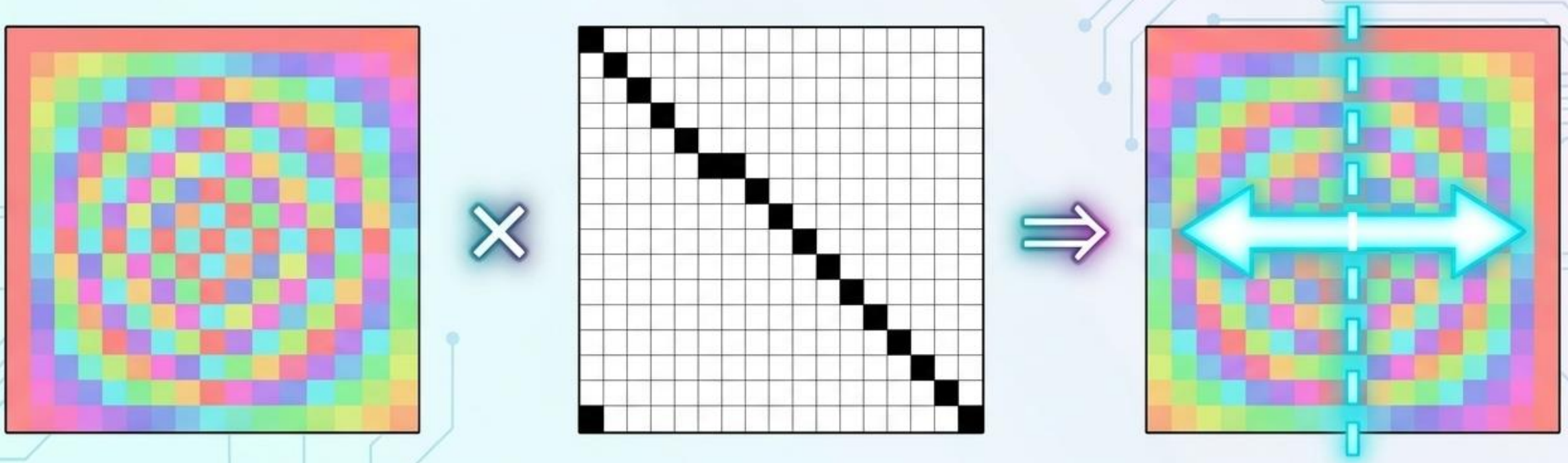
$RX \neq XR$

Matrix multiplication is non-commutative. The visual outputs of Experiment 1 differ because pre-multiplying a matrix is fundamentally, structurally different from post-multiplying it.

The Setup: To understand why QFT and NOT interact the way they do, we must run a new experiment. Let's isolate the mechanical effects of the X gate on a completely random matrix R.

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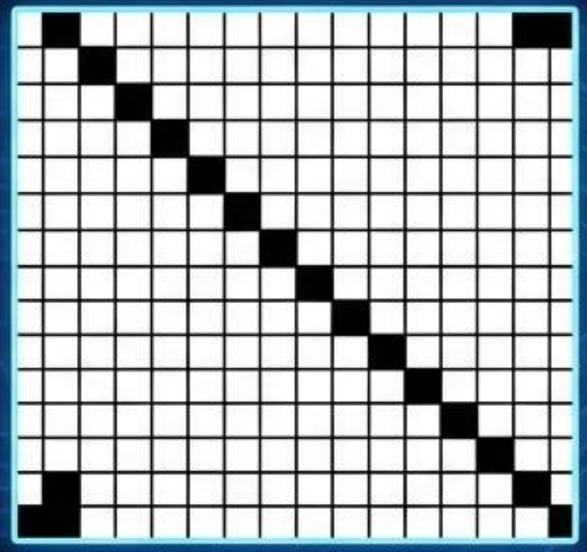
Lab Experiment 2A: Post-Multiplying by X



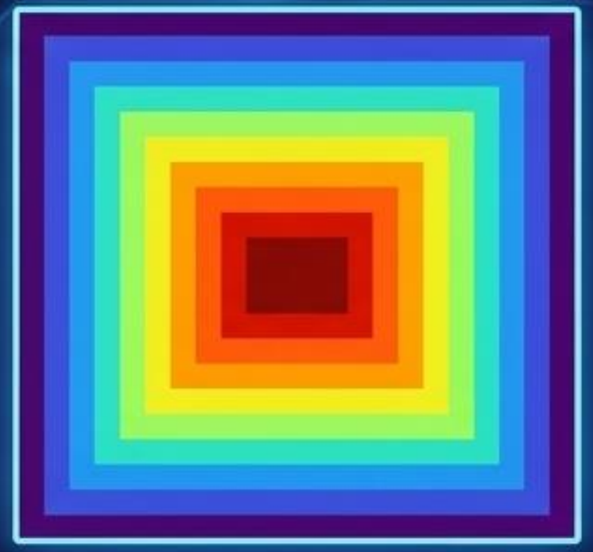
Operation: Right/Post-Multiplication (R ∘ X)
The Mechanical Effect: Applying the X gate after a matrix visually flips the columns of that matrix around its vertical center. Let's not increase a matrix visually flips the columns of that matrix around its vertical center.
Math Equivalent: fliplr(R) (Left-to-Right Flip)

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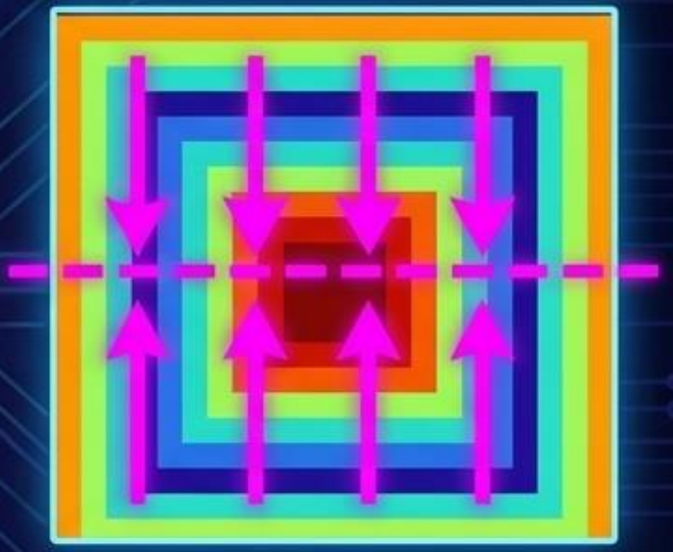
Lab Experiment 2B: Pre-Multiplying by X



X Gate



Random Matrix R



Operation: Left/Pre-Multiplication (X ∘ R)
The Mechanical Effect: Applying the X gate before a matrix visually flips the rows of that matrix around its horizontal center.
Math Equivalent: flipud(R) (Upside-Down Flip)

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The X-Gate Flipping Diagnostic

Post-Multiply (RX)	Pre-Multiply (XR)
The gate operates on the right.	The gate operates on the left.
	
Column Flip (Left-to-Right)	Row Flip (Upside-Down)

Key Insight: By visualizing the matrices, we eliminate the need to memorize arbitrary algebraic rules. The visual behavior perfectly maps to the mechanical truth of the operation.

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Connecting the Experiments

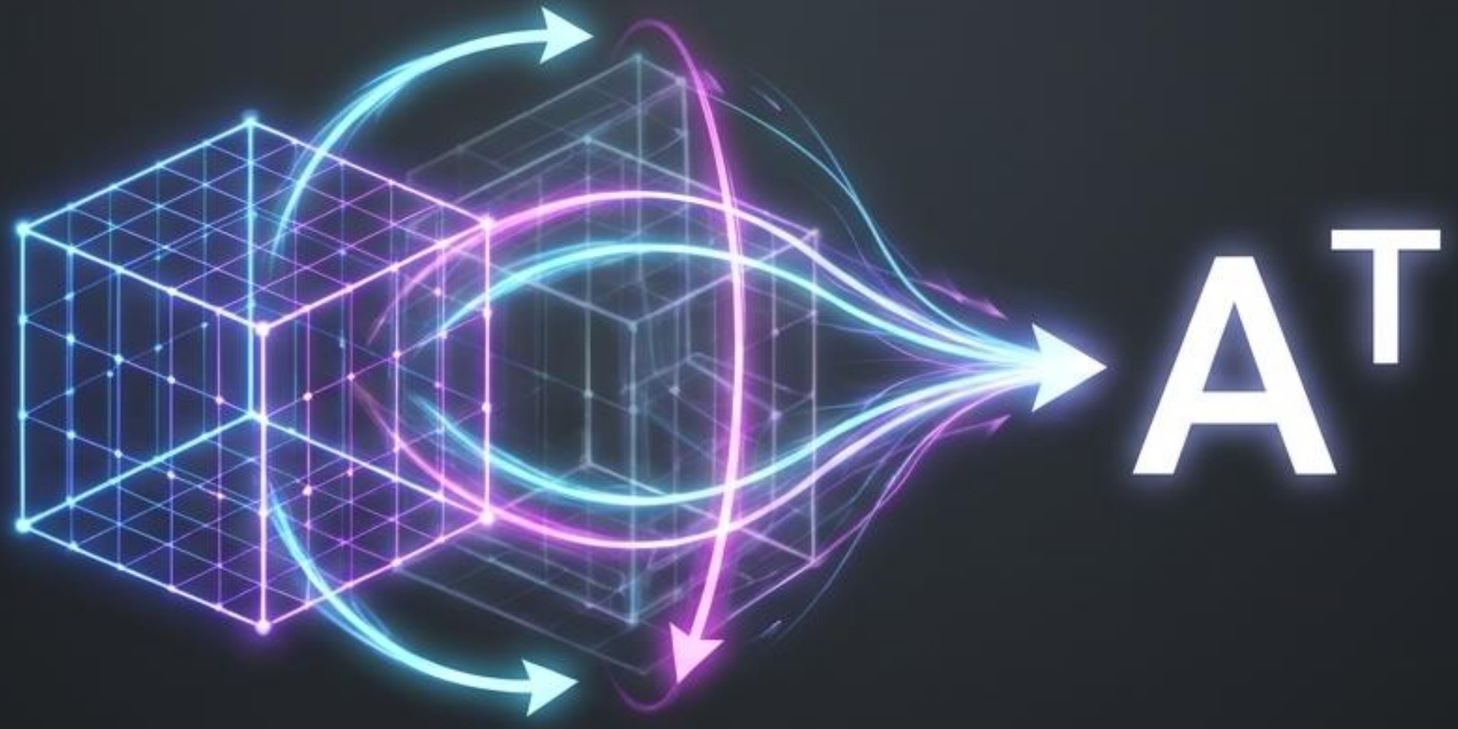
$QFT^2 \circ X$ → Add 1 → Post-Multiply Col Flip

Because QFT^2 is inherently tied to these geometric permutations, the Successor (Add 1) and Predecessor (Subtract 1) operations are fundamentally intertwined.

The Insight: Each rotation is effectively the flip upside-down AND flip left-to-right of the other. The differing directions are simply the mathematical manifestations of horizontal vs. vertical matrix flips.

$X \circ QFT^2$ → Subtract 1 → Pre-Multiply Row Flip

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- **The Observation:** Successor and Predecessor are symmetric along the anti-diagonal.
- **The Geometric Truth:** Flipping a matrix both upside-down and left-to-right is equivalent to its transpose.
- **The Mathematical Theorem:** Therefore, Successor and Predecessor are transposes of one another.

Our visual flipping experiments seamlessly confirm a **deeper theorem of linear algebra:** Transposing inverts permutations.

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The Value of Experimental Mathematics



Suggest

Interactive visuals allow us to spot patterns and suggest tentative mathematical facts instantly, bypassing cognitive overload.



Confirm

'Calculated trial and error' using varying random matrices rapidly verifies structural behaviors (like row vs. column flips).



Prove

Visual intuition grounds formal rigor. Once we 'see' that a matrix behaves as a transpose, we know exactly what to prove mathematically.

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Open the Lab



q-info.github.io/QAL-Lite

Stop thinking in equations. Start experimenting in visuals.

The Quantum Algorithms Lab (QAL) is open source and runs entirely in your browser.
Play with the matrices, break the algorithms, and see the math for yourself.

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